

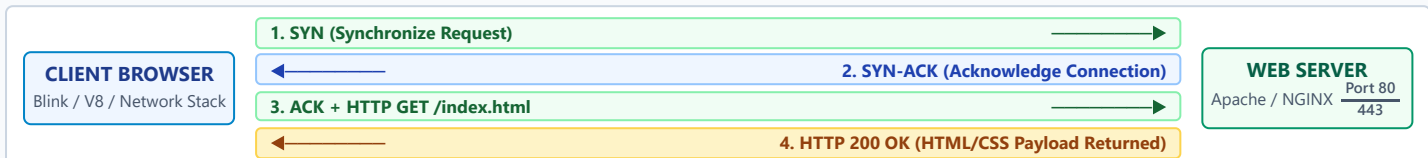
UNIT I: INTERNET & WORLD WIDE WEB ARCHITECTURE (6 Hours)

Core Exam Topics: DNS, HTTP, Client-Server, Search & Casting

FLOWCHART 1.1: COMPLETE 5-TIER HIERARCHICAL DNS RESOLUTION SEQUENCE



FLOWCHART 1.2: TCP 3-WAY HANDSHAKE & HTTP REQUEST-RESPONSE TRANSACTION CYCLE



EXHAUSTIVE COMPARISON: INTERNET VS. WORLD WIDE WEB (WWW)

Feature	The Internet	World Wide Web (WWW)
Nature	Global physical network of interconnected hardware	Software hypermedia collection operating atop Internet
Inception	1969 (ARPANET by US DoD) Flag Day 1983	1989 (Sir Tim Berners-Lee at CERN)
OSI Layer	Lower layers: Physical, Data Link, Network, Transport	Strictly Application Layer (Layer 7)
Protocols	TCP, UDP, IP, IPv4, IPv6, BGP, ICMP, ARP	HTTP, HTTPS, WebSockets, WebRTC
Addressing	IP Address (32-bit IPv4 / 128-bit IPv6)	Uniform Resource Locator (URL / URI)
Governance	IETF, ICANN, IANA	W3C (World Wide Web Consortium), WHATWG

HTTP STATUS CODES & SEARCH ENGINE PIPELINE

HTTP RESPONSE STATUS CLASSIFICATIONS

1xx (Info): 100 Continue, 101 Switching Protocols
2xx (Success): 200 OK, 201 Created, 204 No Content
3xx (Redirect): 301 Moved Permanently, 304 Not Modified
4xx (Client Error): 400 Bad Request, 401 Unauth, 404 Not Found
5xx (Server Error): 500 Internal Error, 502 Bad Gateway, 503 Unavailable

SEARCH ENGINE PIPELINE & WEB-CASTING

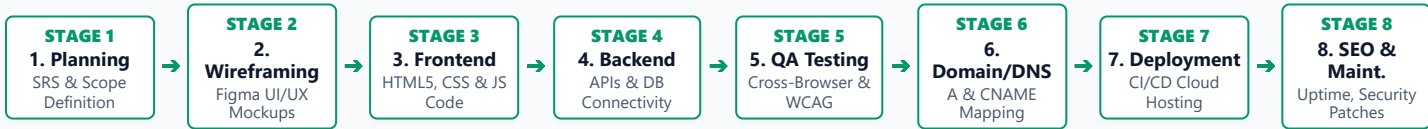
1. Spiders (Crawler Bots) → 2. Parser (Strip & Tokenize) → 3. Indexer (Inverted Index) → 4. Ranking (PageRank / AI)

Web-Casting: Pull (Client pulls data) vs. Push (Server push, WebSockets, SSE); Unicast (1:1) vs. Multicast (1:Many tree replication).

UNIT II: BASICS OF HTML & WEBSITE ENGINEERING (6 Hours)

Core Exam Topics: SDLC, DOM Hierarchy, Image Mapping, Tags & Tables

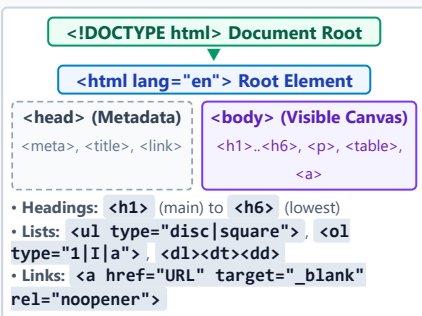
FLOWCHART 2.1: 8-STAGE WEBSITE DEVELOPMENT & PUBLISHING LIFECYCLE (SDLC)



STATIC VS. DYNAMIC WEBSITES

Feature	Static Website	Dynamic Website
Content	Pre-authored, identical for all	Generated on-the-fly via DB/Logic
Database	No database required	Heavily driven by MySQL/Postgres
Tech Stack	HTML, CSS, Vanilla JS	PHP, Node.js, Python, Java
Speed	Ultra-fast CDN disk cache	Server execution & query latency
Updates	Manual per-file editing	Instant via Admin CMS portal
Security	Immune to SQLi / code injection	Needs patches for XSS, SQLi, CSRF

HTML DOCUMENT TREE (DOM HIERARCHY)



CLIENT-SIDE IMAGE MAP SHAPES

```

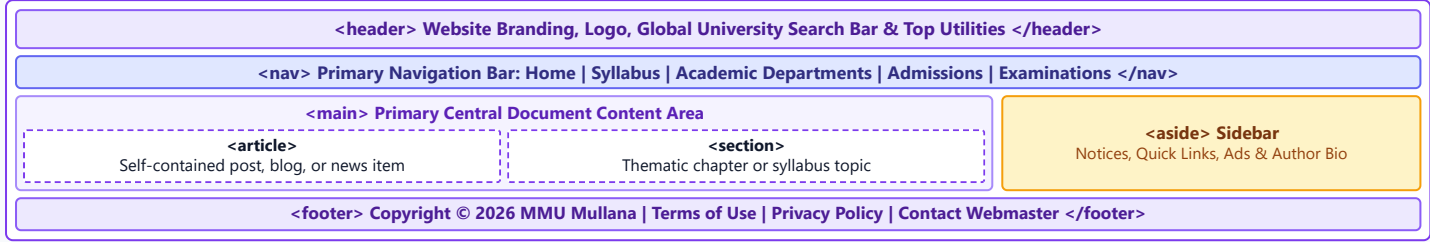
<map name="navmap">
  1. Rect: <area shape="rect"
    coords="x1,y1,x2,y2" href="p1.html">
  2. Circle: <area shape="circle"
    coords="cx,cy,r" href="p2.html">
  3. Poly: <area shape="poly"
    coords="x1,y1,x2,y2,x3,y3"
    href="p3.html">
</map>
```

Exam Note: Legacy <frameset> is deprecated in HTML5; modern standard is <iframe sandbox="...">.

UNIT III: HTML5 & MODERN WEB APPLICATION APIS (6 Hours)

Core Exam Topics: Semantic Blueprint, Audio/Video Multi-Codec, Canvas 2D, Web Storage

FLOWCHART 3.1: HTML5 STRUCTURAL SEMANTIC PAGE LAYOUT BLUEPRINT



WEB STORAGE DECISION TREE

Client Storage Requirement?

Cookies
4KB max
Sent on every HTTP header

localStorage
5-10MB
Permanent across tabs/restarts

sessionStorage
5MB
Cleared when tab closes

API: `setItem(k,v)` | `getItem(k)` | `removeItem(k)` | `clear()`

CANVAS 2D RENDERING PIPELINE

1. DOM Element: `<canvas id="c" width="300" height="150">`

2. Acquire Context: `const ctx = c.getContext("2d");`

3. Render Primitives:

- `ctx.fillRect(x,y,w,h)` (Bar)
- `ctx.arc(x,y,r,0,2*Math.PI)` (Circle)
- `ctx.stroke()` & `ctx.fill()`

MULTIMEDIA & HTML5 FORMS

Multi-Codec Media Fallback:

```
<video controls poster="p.jpg">
  <source src="v.mp4" type="video/mp4">
  <source src="v.webm" type="video/webm">
</video>
```

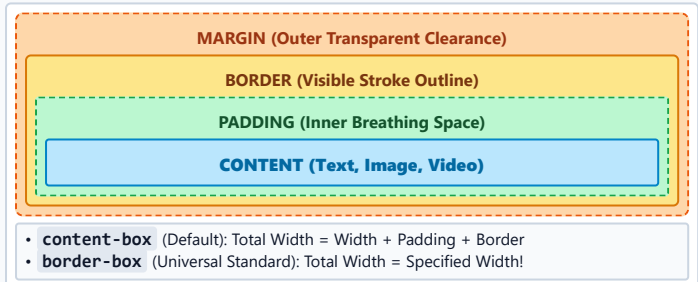
New Form Input Types:
email, number, range, date, color, url, search

Attributes: required, pattern, placeholder, autofocus, min/max

UNIT IV: CASCADING STYLE SHEETS (CSS) ARCHITECTURE (6 Hours)

Core Exam Topics: Box Model, 5 Positioning Schemes, Specificity Cascade, Floats & Tables

FLOWCHART 4.1: CSS BOX MODEL HIERARCHY & SIZING FORMULA



FLOWCHART 4.2: VISUAL COMPARISON OF 5 CSS POSITIONING SCHEMES

static
Normal flow; ignores top/left

relative
Nudged; preserves slot

absolute
Out of flow; nearest parent

fixed
Viewport lock; pins on scroll

sticky
Relative until threshold

- z-index**: Controls overlapping stacking order (applies only to positioned elements with non-static position).
- display**: **block** (full row), **inline** (content width), **inline-block** (accepts w/h), **none** (removed from DOM render).

CSS SPECIFICITY PRIORITY CASCADE

1. !important	Overrides All
2. Inline Style (style="...")	Weight: 1000
3. ID Selector (#header)	Weight: 0100
4. Class / Pseudo (.btn, :hover)	Weight: 0010
5. Element / Tag (p, div, ::after)	Weight: 0001

3 CSS APPLYING METHODS

Method	Syntax / Location	Evaluation
Inline	<code><p style="..."></code>	Worst practice; zero reusability
Internal	<code><style></code> in <code><head></code>	Single page only; no multi-page cache
External	<code><link rel="stylesheet"></code>	Industry Standard: Browser caching + decoupling

FLOATS, CLEARFIX & TABLE RESET

Micro clearfix Hack:

```
.clearfix::after { content: ""; display: table; clear: both; }
```

Professional Table Styling:

```
table { border-collapse: collapse; width: 100%; }
th, td { border: 1px solid #cbd5e1; padding: 6px; }
tr:nth-child(even) { background: #f8fafc; }
```